

Ye Yuan

Ph.D. Candidate, Computer Science, McGill University, Mila - Quebec AI Institute
Visiting Scholar, Machine Learning, MBZUAI
DAAD AINeT Fellowship, BMO Responsible AI Fellowship

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Research Interests

Areas: Large Language Models, Score-Based Generative Models, Offline Black-Box Optimization, Knowledge-Centric Natural Language Processing, Retrieval-Augmented Generation, Autonomous Agents.

My research centers on developing *large language models* and *score-based generative models* to advance intelligent systems, with applications in (i) addressing challenges in offline black-box optimization, (ii) enabling knowledge-centric NLP tasks such as automatic knowledge base construction, and (iii) improving the efficiency and efficacy of AI systems for real-world applications. My work is supported by multiple fellowships and has been published at top venues including NeurIPS, ICLR, EMNLP, and TMLR. **Representative contributions include authoring a comprehensive survey on offline black-box optimization with Turing Award Winner Prof. Yoshua Bengio and presenting an EMNLP 2024 oral paper (top 3%),** complemented by industrial research experiences at Microsoft Research, RBC Borealis, and Samsung Research America.

Academic Experience

- Oct. 2025 - **MBZUAI - Mohamed bin Zayed University of Artificial Intelligence, Abu Dhabi, United Arab Emirates.**
Present Visiting Scholar, Machine Learning.
Faculty Host: [Associate Vice President for Research, Prof. Steve Liu.](#)
- Jan. 2023 - **McGill University & Mila - Quebec AI Institute, Montreal, Canada.**
Present Doctor of Philosophy (Ph.D.), Computer Science.
CGPA: 3.89 / 4.00.
Supervisor: [Prof. Steve Liu.](#)
Supervisory Committee: [Prof. Adriana Romero-Soriano](#); [Prof. Gintare Karolina Dziugaite.](#)
- Sep. 2019 - **McGill University & Mila - Quebec AI Institute, Montreal, Canada.**
Dec. 2022 Bachelor of Science (B.Sc.), Honours Computer Science.
CGPA: 3.95 / 4.00.
First Class Honours; Dean's Honour List (Top 10%); Recipient of Faculty of Science Scholarship.

Industrial Experience

- May. 2025 - **RBC Borealis, Toronto, Canada.**
Oct. 2025 **Research Intern.**
Manager: [Dr. Eric He](#); Project Mentors: [Dr. Mohammad Amin Shabani](#); [Dr. Siqi Liu.](#)
- Led two concurrent research projects that resulted in two conference paper submissions and one accepted NeurIPS workshop paper within a single internship term.
 - Proposed an embedding-based context-aware reranking algorithm leveraging structural information and hybrid attention, improving both accuracy and efficiency.
 - Developed an LLM-based agentic framework for query-focused table summarization, achieving higher summarization quality while ensuring privacy compliance by avoiding raw data exposure.

Jan. 2025 - **Samsung Research America**, Mountain View, United States.

Present **Student Researcher** (Remote).

Manager: [Dr. Charlie Zhang](#).

- Designed entity extraction strategies to automatically construct domain-specific knowledge bases, enabling scalable knowledge graph creation for telecommunications data.

Feb. 2023 - **Microsoft Research**, Cambridge, United Kingdom.

Present **Student Researcher** (Remote).

Manager: [Yordan Zaykov](#); Project Mentors: [Dr. Bhaskar Mitra](#); [Dr. Tom Minka](#).

- Introduced and formalized the structured entity extraction as a new paradigm within information extraction.
- Proposed a new evaluation metric (AESOP) and its variants for structured entity extraction, better aligning evaluation with human judgment.
- Developed a multi-stage decoding strategy that improved extraction effectiveness and efficiency.
- Analyzed the learning dynamics of score-based generative models and LLM-based discriminative models in structured entity linking tasks through a series of controlled experiments.

Apr. 2023 - **Noah's Ark Lab**, Montreal, Canada.

Mar. 2025 **Research Intern**.

Manager: [Dr. Alex Chen](#).

- Recognized with multiple awards, including **Outstanding Contribution Award (R&D Peripheral Fields)**, **Overseas Business Contribution Award**, and **President's Spot Award (Canada Research Institute)**.
- Adapted soft prompting techniques to address distribution shift, enabling a single model to generalize across multiple tasks.
- Explored model compression methods (e.g., LoRA-pruning, Mixture-of-Depth) to reduce computation cost for edge deployment.
- Investigated state-space language models and hybrid architectures with transformers for improved efficiency and scalability.

Publications

Within the first three years of my Ph.D., I produced around 10 peer-reviewed publications, 7 as (co-)first author, across top-tier venues such as NeurIPS, ICLR, EMNLP, and TMLR. By October 2025, my work had attracted 430+ citations (h-index 8) on Google Scholar, and my open-source contributions were recognized with 120+ stars on GitHub.

Submissions / Preprints

- S8 **Ye Yuan**, Mohammad Amin Shabani, Siqi Liu. “*FACTS: Table Summarization via Offline Template Generation with Agentic Workflows*”. Under Review.
- S7 **Ye Yuan**, Mohammad Amin Shabani, Siqi Liu. “*Embedding-Based Context-Aware Reranker*”. Under Review.
- S6 **Ye Yuan***, Can Chen*, Zipeng Sun, Dinghuai Zhang, Christopher Pal, Xue Liu. “*Diffusion Large Language Models for Black-Box Optimization*”. Under Review.
- S5 Linfeng Du, **Ye Yuan**, Fuyuan Lyu, Emiliano Penaloza, Xiuying Chen, Zipeng Sun, Laurent Charlin, Xue Liu, Haolun Wu. “*Optimizing User Profiles via Contextual Bandits for Retrieval-Augmented LLM Personalization*”. Under Review.
- S4 Minsu Kim*, Jiayao Gu*, **Ye Yuan**, Taeyoung Yun, Zixuan Liu, Yoshua Bengio, and Can Chen. “*Offline Model-Based Optimization: Comprehensive Review*”. arXiv preprint arXiv:2503.17286.

- S3 Hao Zhou, Chengming Hu, Dun Yuan, **Ye Yuan**, Di Wu, Xue Liu, and Charlie Zhang. “*Large Language Model (LLM)-Enabled In-Context Learning for Wireless Network Optimization: A Case Study of Power Control*”. arXiv preprint arXiv:2408.00214.
- S2 Shangyu Wu, Ying Xiong, Yufei Cui, Haolun Wu, Can Chen, **Ye Yuan**, Lianming Huang, Xue Liu, Tei-Wei Kuo, Nan Guan, Chun Jason Xue. “*Retrieval-Augmented Generation for Natural Language Processing: A Survey*”. arXiv preprint arXiv:2407.13193.
- S1 Shuhao Zheng, Yanxi Lin, Yang Yu, **Ye Yuan**, Yongzheng Jia, Xue Liu. “*FIAT: Fine-grained Information Audit for Trustless Transborder Data Flow*”. arXiv preprint arXiv:2209.11451.

Journal Articles

- J4 **Ye Yuan**^{*}, Youyuan Zhang^{*}, Can Chen, Haolun Wu, Zixuan Li, Jianmo Li, James J. Clark, and Xue Liu. TMLR “*Design Editing for Offline Model-based Optimization*”. Transactions on Machine Learning Research, 2025.
- J3 Hao Zhou, Chengming Hu, Dun Yuan, **Ye Yuan**, Di Wu, Xi Chen, Hina Tabassum, and Xue Liu. “*Large Language Models for Wireless Networks: An Overview from the Prompt Engineering Perspective*”. IEEE WCM IF: 11.5 Wireless Communications, 2024.
- J2 Hao Zhou, Chengming Hu, Dun Yuan, **Ye Yuan**, Di Wu, Xue Liu, Zhu Han, and Jianzhong Zhang. “*Generative AI as a Service in 6G Edge-Cloud: Generation Task Offloading by In-Context Learning*”. IEEE WCL IF: 5.5 Wireless Communications Letters, 2024.
- J1 Hao Zhou, Chengming Hu, **Ye Yuan**, Yufei Cui, Yili Jin, Can Chen, Haolun Wu, Dun Yuan, Li Jiang, Di Wu, Xue Liu, Charlie Zhang, Xianbin Wang, and Jiangchuan Liu. “*Large Language Model (LLM) for Telecommunications: A Comprehensive Survey on Principles, Key Techniques, and Opportunities*”. IEEE COMST IF: 46.7 Communications Surveys and Tutorials, 2024.

Conference Proceedings

- C4 **Ye Yuan**, Haolun Wu, Hao Zhou, Xue Liu, Hao Chen, Yan Xin, Jianzhong (Charlie) Zhang. “*Understanding 6G through Language Models: A Case Study on LLM-aided Structured Entity Extraction in Telecom Domain*”. GlobeCom Oral, 39.6% In the 2025 IEEE Global Communications Conference. Taipei, Taiwan. Dec. 2025.
- C3 **Ye Yuan**^{*}, Can Chen^{*}, Christopher Pal, Xue Liu. “*ParetoFlow: Guided Flows in Multi-Objective Optimization*”. In the 13th International Conference on Learning Representations. Singapore EXPO, Singapore. Apr. 2025. Poster, 32.1%
- C2 **Ye Yuan**^{*}, Haolun Wu^{*}, Liana Mikaelyan, Alexander Meulemans, Xue Liu, James Hensman, and Bhaskar Mitra. “*Learning to Extract Structured Entities Using Language Models*”. In the 19th Conference on Empirical Methods in Natural Language Processing. Miami, Florida, USA. Nov. 2024. EMNLP Oral, Top 3%
- C1 **Ye Yuan**^{*}, Can Chen^{*}, Zixuan Liu, Willie Neiswanger, Xue Liu. “*Importance-aware Co-teaching for Offline Model-based Optimization*”. In the Advances in Neural Information Processing Systems 36. New Orleans, Louisiana, USA. Dec. 2023. NeurIPS Poster, 26.1%

Workshops

- W3 **Ye Yuan**, Mohammad Amin Shabani, Siqi Liu. “*FACTS: Fast, Accurate, and Privacy-Compliant Table Summarization via Offline Template Generation*”. In NeurIPS 2025 Workshop on Generative AI in Finance. NeurIPS
- W2 Hao Zhou, Chengming Hu, Dun Yuan, **Ye Yuan**, Di Wu, Xue Liu, Jianzhong Charlie Zhang. “*Prompting Wireless Networks: Reinforced In-Context Learning for Power Control*”. In ICML 2025 Workshop on Machine Learning for Wireless Communication and Networks. ICML
- W1 **Ye Yuan**^{*}, Youyuan Zhang^{*}, Can Chen, Haolun Wu, Melody Zixuan Li, Jianmo Li, James J. Clark, Xue Liu. “*Design Editing for Offline Model-based Optimization*”. In ICLR 2025 Workshop on Deep Generative Model in Machine Learning: Theory, Principle and Efficacy. ICLR

Grants & Fellowships

Total funding: \$331,000. Total share of funding: \$140,500.

I led or contributed to the work of the following grant and fellowship proposals.

- 2025 DAAD AINeT Fellowship (\$5,000, share 100%) (1 of 18 AInet Fellows in Natural Language Processing)
Funding Body: DAAD
- 2024 Samsung Global Research Outreach Program (\$210,000, share 25%) (1 of 11 Recipients Globally)
Lead PI: Prof. Steve Liu
Title: LLM-Driven Efficient and Flexible Network Management for NextG Cellular Communications
Funding Body: Samsung Research
- 2023 MSR-Mila Collaboration Grant (\$66,000, share 50%)
Lead PI: Prof. Steve Liu; Industrial Mentor: Dr. Bhaskar Mitra
Title: SNAKE: Structured Neural Attention over Knowledge base Entities
Funding Body: Microsoft Research
- 2023 BMO Responsible AI Fellowship (\$50,000, share 100%)
Funding Body: Bank of Montreal (BMO)

Honors & Awards

Total award amount: \$25,450.

- 2025 AInet Fellows in Natural Language Processing, DAAD
- 2025 Graduate Excellence Award, McGill University
- 2025 Student Travel Award, International Conference on Learning Representations (ICLR)
- 2023-2025 Responsible AI Senior Scholar, Bank of Montreal
- 2023-2025 Faculty of Science Graduate Scholarship, McGill University
- 2024 Outstanding Contribution Award in R&D Peripheral Fields, Noah's Ark Lab Canada
- 2024 Overseas Business Contribution Award, Noah's Ark Lab Canada
- 2024 Canada Research Institute President's Spot Award, Noah's Ark Lab Canada
- 2023 Scholar Award, Neural Information Processing Systems (NeurIPS)
- 2022 Graduate Excellence Award, McGill University
- 2022 Graduated as First Class Honours, McGill University
- 2020-2021 Tomlinson Undergraduate Award, McGill University
- 2019-2022 Dean's Honour List, McGill University

Media

I actively share and communicate research on social media, with 1,600+ followers across LinkedIn and X.

- 2024 Oral Presentation of Learning to Extract Structured Entities Using Language Models. EMNLP Recording.
- 2024 Large Language Model (LLM) for Telecommunications: A Comprehensive Survey on Principles, Key Techniques, and Opportunities. IEEE COMST Post.
- 2023 Importance-Aware Co-Teaching for Offline Model-Based Optimization. Mila Blog.

Invited Talk, Teaching & Mentorship

Invited Talk

- "Generative VS. Discriminative Approaches for Entity Linking.", Microsoft Research.

Host: [Yordan Zaykov](#). May. 2025.

- “*Offline Model-Based Optimization*”, Beijing Academy of Artificial Intelligence.

Host: [Dr. HongJiang Zhang](#). Jul. 2024.

- “*Offline Model-Based Optimization*”, ByteDance AI Lab.

Host: [Dr. Hang Li](#). Jun. 2024.

○ Teaching

- “*COMP 202: Foundations of Programming*”, Teaching Assistant, McGill University.

Worked as a Teaching Assistant for two semesters at McGill for COMP 202: Foundations of Programming. This course is an introduction to computer programming in a high level language. I worked with [Prof. Faten M'hiri](#) for this course.

- “*COMP 250: Introduction to Computer Science.*”, Teaching Assistant, McGill University.

Worked as a Teaching Assistant for two semesters at McGill for COMP 250: Introduction to Computer Science. This course is about Mathematical tools, Datastructures, Recursive and non-recursive algorithms, Abstract data types and Object oriented programming in Java. I worked with [Prof. Giulia Alberini](#) for this course.

○ Mentorship

- “*Mila Industry Mentorship Program for Machine Learning Internships*”, Mentor, Mila - Quebec AI Institute.

Mentored two professional Master students for their internship projects. It was my fortunate to work with Arvind Nair for mentoring his internship at Bell Canada with Prof. Dhanya Sridhar. It was my fortunate to work with Paul Kelendji for mentoring his internship at DeepRiver with Prof. Bang Liu.

Professional & Community Service

○ Conferences Reviewer

- ICLR 2026
- AISTATS 2026
- NeurIPS 2025
- ACL Rolling Review 2025 (ACL, EMNLP)
- ICML 2025
- ICLR 2025
- AISTATS 2025
- NeurIPS 2024

○ Journals Reviewer

- TMLR
- ACM TOIS

○ Workshops Reviewer

- ICLR DeLTa 2025
- AAAI Undergrad Consortium 2025
- AAAI KnowFM 2025
- NeurIPS SafeGenAi 2024
- ICML FM-Wild 2024

○ Community Service

- Mentor and Judge for McHacks Hackthon 9, 10, 11

Extracurricular Activities

- One of my biggest hobby is playing **Xiangqi** (Chinese Chess), which is a strategy board game and is the most popular board game in China. The game represents a battle between two armies, with the primary object being to checkmate the enemy's general. It is my fortunate to be mentored by the World Xiangqi Federation international master Bo Peng.
- When I was at junior high and senior high school, I was a member of the school's **basketball varsity**. We won the district's championship once, and the International Schools Basketball Cup Championship once.
- Before studying computer science, I was quite interested in **aerodynamics for plane design**. I received the China National First Tier Project award once, and Beijing Municipal First Tier Project award once.